The supply chain effects of US tariffs

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The United States of America imposed tariffs on every country in 2025, with a baseline of 10%, ranging up to 50% for the around 60 countries that received higher than 10% tariffs . This presents a great challenge to global supply chain networks. Although supply chain networks in their entirety are considered as scale-free networks, when aggregating them to larger regional levels, they more closely represent a small-world network. In case of global trade policy, this consideration is more appropriate, due to the fact that these policies don't affect links at the end nodes, which represent companies, rather the connection between larger hubs.

This topic is well researched, regarding the negative effects on the links between the trading hubs of the imposing country and their recipient, and how this can reshape the network making changes in various levels of the network (Eugster et al., 2022; Alves et al., 2018). Furthermore, the increase in tariffs has a negative impact on the number of participants in the global value chain (Eugster et al., 2022; Piccardi & Tajoli, 2018). Another well discussed topic is the harmful effects of tariffs on domestic producers that rely on imported inputs (Amiti et al., 2019; Eugster et al., 2022). This can have a larger impact on higher complexity goods, due to them relying on more diverse inputs, some of which are also complexed with few countries specialized in them (Piccardi & Tajoli, 2018).

An interesting topic, which hasn't received much attention, is how these tariffs affect average path length in supply chain networks, and how product complexity plays a role in this aspect. Companies often create or expand offshore branches to circumvent imposed tariffs. Navigating this procedure is relatively easy, when only a single country is impacted by tariffs. Here the challenge is that several countries are hit with greatly differing tariffs on most product types from a lucrative, large market.

Several stakeholders can benefit from taking the appropriate action. Companies can gain competitive advantages by making the right choices. Countries with low tariff rates may be able to attract foreign investment and may move up on the complexity ladder. Other countries can better position themselves if they establish new or improve existing logistical hubs, based on the emerging networks, aiming to reduce the potential increase of the average path length. On a more local scale politicians and analysts can identify key areas where lay-offs can happen and take preventative actions to limit the increase of unemployment in vulnerable areas.

References

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